Holy Angel University
School of Computing
Computer Science Department

Intelligent System

Energy Demand Forecasting + RL Load Scheduling (Time Series + RL)

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Submitted by:
Alvarez, Rainier
Esguerra, Keane
Morata, Nick Christian
Muli, Icent Kurt

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Submitted to:
MA. LOUELLA M SALENGA

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ABSTRACT

This project shows an intelligent system that uses computer vision, natural language processing, and reinforcement learning to mimic how people make decisions based on traffic signs. To sort traffic signs from pictures, a Convolutional Neural Network or known as CNN is used. Then a Reinforcement Learning agent that works in a simulated grid environment uses the predicted sign as input. There is also a small NLP or natural language processing that is a part that works with text labels that are related to the traffic signs. The point of this project is to show how to combine different AI methods into one pipeline. We will use classification accuracy, the macro F1 score, the cumulative reward and the success rate to test the system. There is also talk about ethical issues like safety limits, dataset bias and the chance of misuse.

INTRODUCTION

Smart systems, especially for autonomous vehicles, driver-assistance systems, and robots, rely on the essential component of traffic sign recognition. The system should not only recognize the sign, but the machine should also act according to the knowledge gained from it. This involves a combination of different AI techniques, including deep learning, reinforcement learning, and natural language processing.

This project involves the creation of a smart system for the classification of traffic signs using a Convolutional Neural Network (CNN). The system will then use the classified sign as input as a reinforcement learning (RL) agent running a simulated environment. The system will incorporate a small natural language processing (NLP) module for the processing of text labels used for traffic signs.

The system involves image classification, reinforcement learning within a simulated environment, and natural language processing. The system will use Python as the programming language, along with deep learning libraries for the implementation of the system. The system's success can be measured by the accuracy, macro-F1, reinforcement learning reward, and the reinforcement of learning success rate. The system's limitations are the computing resources, as the system must run within a limited runtime constraint.

RELATED WORK

Deep learning, specifically Convolutional Neural Networks (CNNs), is the dominant technique for traffic sign image classification. The German Traffic Sign Recognition Benchmark (GTSRB) is a common benchmark for the evaluation of traffic sign image classification systems.

Reinforcement learning involves the use of a learning agent that uses trial and error to learn from the environment, which involves the use of rewards or penalties. The use of reinforcement learning with computer vision allows the system to make decisions based on the images captured by the camera.

Natural language processing is a technique used for the processing of text labels, which can improve the system's understanding of the images captured by the camera.

Recent studies have shown that the use of a combination of different AI techniques within a single system result in a robust system. This project combines the use of convolutional neural networks, reinforcement learning, and natural language processing within a single system.

DATASET PLAN

The dataset for the project is based on the German Traffic Sign Recognition Benchmark (GTSRB). The dataset is a collection of images of traffic signs labeled appropriately. The dataset is commonly used for machine learning purposes.

The dataset is available for public use for academic purposes. The dataset does not contain any personal data or any data that can be used to identify individuals. Therefore, no issue of privacy or consent is applicable to the dataset. The dataset contains images of multiple signs taken under different lighting conditions and at different angles. This is advantageous for training a convolutional neural network. To avoid data leakage, the dataset will be split into training, validation, and test sets. The images will be resized before training.

METHOD AND SYSTEM DESIGN

The core component of the system will be a Convolutional Neural Network for the classification of the signs. A simple Convolutional Neural Network architecture will be implemented from scratch to show the understanding of the concepts of deep learning.

The reinforcement learning component will be implemented as a Q-learning algorithm. The RL component will be implemented as an agent that will be required to perform a task based on the classified signs.

A Natural Language Processing component will be implemented to handle the text-based labels for the signs. This will enable the system to map the signs to actions that the reinforcement learning component will be able to understand. The minimum viable product will comprise a Convolutional Neural Network classifier, a reinforcement learning component, and a Natural Language Processing component.

SUCCESS METRICS AND CONSTRAINTS

The Convolutional Neural Network classifier will be evaluated based on accuracy and macro F1. A confusion matrix will also be implemented to evaluate the classification errors. The reinforcement learning component will be evaluated based on cumulative reward, success rate, and learning curves.

The Natural Language Processing component will be evaluated based on the accuracy of the mapping. The constraints will comprise training time, available GPU resources, and the requirement for the entire system to execute within the allowed time.

ETHICS RISK REGISTER

There is a risk that the system might be abused for the purpose of the project. The project will be implemented for educational purposes only. Therefore, a disclaimer will be included to show that the system is for educational purposes. There is a risk that the data might be biased. The data might not comprise all the cases that might be encountered in the real world. This will be addressed in the model card.

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